
CAS MA394 APPLIED ABSTRACT ALGEBRA

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1 Lecture 1 – 1/20

1.1 1.2 Sets and Equivalence Relations

Definition 1.1. A **set** is a well-defined collection of objects; that is, it is defined in such a manner that we can determine for any given object whether or not it belongs to the set. The objects that belong to a set are called its **elements** or **members**. We denote them with capital letters such as A or X .

- If a is an element of a set A , we write $a \in A$.
- The set without any elements is the **empty set**, denoted $\emptyset$ or $\{\}$.
- If A and B are sets and every element of A is in B , then A is a **subset** of B , denoted $A \subseteq B$.
- If $A \subset B$ and $B \subset A$, then $A = B$.
- If $A \subset B$ and $A \neq B$, then A is a **proper subset** of B , denoted $A \subset B$.

Some standard sets include:

- $\mathbb{N} = \{1, 2, 3, \dots\}$, the set of natural numbers (positive integers).
- $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$, the set of integers.
- $\mathbb{Q} = \left\{\frac{a}{b} \mid a, b \in \mathbb{Z}, b \neq 0\right\}$, the set of rational numbers.
- $\mathbb{R} = (-\infty, \infty)$, the set of real numbers.
- $\mathbb{C} = \{a + bi \mid a, b \in \mathbb{R}, i^2 = -1\}$, the set of complex numbers.

We can also build new sets from old sets.

- The **union** of two sets A and B is the set of elements that are in A or B (or both), denoted $A \cup B$.

$$A \cup B = \{x : x \in A \text{ or } x \in B\}.$$

- The **intersection** of two sets A and B is the set of elements that are in both A and B , denoted $A \cap B$.

$$A \cap B = \{x : x \in A \text{ and } x \in B\}.$$

- The **complement** of a set A (with respect to a universal set U) is the set of elements in U that are not in A , denoted A' , A^c , $\bar{A}$.

$$A' = \{x : x \in U \text{ and } x \notin A\}.$$

- The **difference** of two sets A and B is the set of elements that are in A but not in B , denoted

$A \setminus B$ or $A - B$.

$$A \setminus B = A \cap B' = \{x : x \in A \text{ and } x \notin B\}.$$

Definition 1.2. De Morgan's Laws state that for any two sets A and B ,

$$(A \cup B)' = A' \cap B'$$

$$(A \cap B)' = A' \cup B'.$$

Definition 1.3. The **Cartesian product** of two sets A and B is the set of ordered pairs (a, b) where $a \in A$ and $b \in B$, denoted $A \times B$.

$$A \times B = \{(a, b) : a \in A, b \in B\}.$$

An example of a Cartesian product is $\mathbb{R}^2 = \mathbb{R} \times \mathbb{R} = \{(x, y) : x, y \in \mathbb{R}\}$, the set of all points in the plane.

Definition 1.4. A **relation** R from a set A to a set B is a subset of the Cartesian product $A \times B$.

Example 1.5. Let $A = \{0, 1, 2, 3\}$, $B = \{0, 1, 2, 3\}$. Then $A \times B = \{(a, b) : a, b \in A\}$. If we say that R is the set $\{(0, 1), (0, 2), (0, 3), (1, 2), (1, 3), (2, 3)\} \subset A \times B$, then R is a relation from A to B . The relation R can be interpreted as "is less than", since for each $(a, b) \in R$, we have $a < b$.

Definition 1.6. We define a mapping or function $f \subset A \times B$ from a set A to a set B as a special type of relation where each $a \in A$ has a unique $b \in B$ such that $(a, b) \in f$. We denote this by $f : A \rightarrow B$ or $A \xrightarrow{f} B$. The set A is called the **domain** of f and $f(A) = \{f(a) : a \in A\} \subset B$ is called the range or **image** of f .

2 Lecture 2 – 1/21

Definition 2.1. If $f : A \rightarrow B$ is a map and the image of f is B , i.e., $f(A) = B$, then f is called **onto** or **surjective**. In other words, if there exists an $a \in A$ for every $b \in B$ such that $f(a) = b$, then f is onto.

Definition 2.2. A map is **one-to-one** or **injective** if $a_1 \neq a_2$ implies $f(a_1) \neq f(a_2)$. Equivalently, a function is one-to-one if $f(a_1) = f(a_2)$ implies $a_1 = a_2$.

Definition 2.3. A map $f : A \rightarrow B$ is a **bijection** if it is both one-to-one and onto.

Definition 2.4. Given two functions, we can construct a new function by using the range of the first function as the domain of the second function. Let $f : A \rightarrow B$ and $g : B \rightarrow C$ be mappings. Define a new map $g \circ f : A \rightarrow C$ called the **composition** of f and g from A to C , by $g \circ f(x) = g(f(x))$.

- If S is any set, we use id_S or id to denote the **identity mapping** from S to itself, or $id(s) = s$ for all $s \in S$.
- A map $g : B \rightarrow A$ is an **inverse mapping** of $f : A \rightarrow B$ if $g \circ f = id_A$ and $f \circ g = id_B$; in other words, the inverse function of a function simply "undoes" the function.
- A map is said to be **invertible** if it has an inverse. We write f^{-1} to denote the inverse of f .

3 Lecture 3 – 1/22

Definition 3.1. An **equivalence relation** on a set X is a relation $R \subset X \times X$ such that

- $(x, x) \in R$ for all $x \in X$ (reflexive property),
- $(x, y) \in R$ implies $(y, x) \in R$ (symmetric property),
- $(x, y) \in R$ and $(y, z) \in R$ imply $(x, z) \in R$ (transitive property).

Given an equivalence relation R on a set X , we usually write $x \sim y$ instead of $(x, y) \in R$. If the equivalence relation already has an associated notation such as $=, \cong, \equiv$, we use that notation instead.

Example 3.2. Let $X = \{1, 2, 3, 4\}$. A set R that is reflexive, symmetric, and transitive (and thus an equivalence relation) is $R = \{(1, 1), (2, 2), (3, 3), (4, 4), (1, 2), (2, 1), (1, 3), (3, 1), (2, 3), (3, 2)\}$.

Definition 3.3. A **partition** P of a set X is a collection of nonempty sets $X_1, X_2, \dots, X_n$ such that $X_i \cap X_j = \emptyset$ and $\bigcup_i X_i = X$. Let $\sim$ be an equivalence relation on a set X and let $x \in X$. Then $[x] = \{y \in X : x \sim y\}$ is called the **equivalence class** of x .

Example 3.4. Let $\mathbb{Z}$ be partitioned into $A = \{\text{even integers}\}$ and $B = \{\text{odd integers}\}$. $A \cap B = \emptyset$ and $A \cup B = \mathbb{Z} \implies A, B$ partition the integers. Define a relation $\sim$ on $\mathbb{Z}$ where $x \sim y \iff x, y$ both even or both odd. Every $a \in A$ is equivalent to every other $a' \in A$ and every $b \in B$ is equivalent to every other $b' \in B$.

4 Lecture 4 – 1/27

4.1 2.1 Mathematical Induction

Definition 4.1. First Principle of Mathematical Induction: Let $S(n)$ be a statement about integers for $n \in \mathbb{N}$ and suppose $S(n_0)$ is true for some integer $n_0 \in \mathbb{N}$. If for all integers k with $k \geq n_0$, $S(k)$ implies that $S(k+1)$ is true, then $S(n)$ is true for all integers $n \geq n_0$.

Example 4.2. For all $n \in \mathbb{N}$, $1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}$.

The base case is $n = 1$; then $1 = \frac{1(1+1)}{2} = 1$, so the base case holds.

Inductive hypothesis: Assume $1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}$ for some n .

Inductive step: We want to show that the statement is true for $n+1$. Consider $1 + 2 + 3 + \cdots + n + (n+1) = \frac{n(n+1)}{2} + (n+1) = (n+1)(\frac{n}{2} + 1) = (n+1)(\frac{n+2}{2}) = \frac{(n+1)(n+1+1)}{2}$.

Example 4.3. Prove that

$$\frac{1}{2} + \frac{1}{6} + \cdots + \frac{1}{n(n+1)} = \frac{n}{n+1}$$

for $n \in \mathbb{N}$.

Base: $n = 1$, then $\frac{1}{1 \cdot 2} = \frac{1}{2}$, so the base case holds.

Inductive hypothesis: Assume $\frac{1}{2} + \frac{1}{6} + \cdots + \frac{1}{n(n+1)} = \frac{n}{n+1}$ for some n .

Inductive step: We want to show that the statement is true for $n+1$. Consider

$$\begin{aligned} \frac{1}{2} + \frac{1}{6} + \cdots + \frac{1}{n(n+1)} + \frac{1}{(n+1)(n+2)} &= \frac{n}{n+1} + \frac{1}{(n+1)(n+2)} \\ &= \frac{n(n+2) + 1}{(n+1)(n+2)} \\ &= \frac{(n+1)^2}{(n+1)(n+2)} \\ &= \frac{n+1}{n+2}. \end{aligned}$$

Therefore the statement is true for all n .

4.2 2.2 The Division Algorithm

Definition 4.4. Let a and b be integers with $b > 0$. Then there exist unique integers q and r such that $a = bq + r$ where $0 \leq r < b$.

- Let a and b be integers. If $b = ak$ for some integer k , we write $a|b$.
- An integer d is called a **common divisor** of a and b if $d|a$ and $d|b$.
- The **greatest common divisor** of a and b is a positive integer d such that d is a common divisor of a and b , and if d' is any other common divisor of a and b , then $d'|d$. We write $d = \gcd(a, b)$.
- We say that two integers a and b are relatively prime if $\gcd(a, b) = 1$.

Proof of the $\gcd$ theorem uses the division algorithm:

$$\begin{aligned}
 a &= q_1b + r_1, 0 \leq r_1 < b \\
 b &= q_2r_1 + r_2, 0 \leq r_2 < r_1, r_1 \neq 0 \\
 r_1 &= q_3r_2 + r_3, 0 \leq r_3 < r_2, r_2 \neq 0 \\
 &\vdots \\
 r_{n-2} &= q_nr_{n-1} + r_n, 0 \leq r_n < r_{n-1}, r_{n-1} \neq 0 \\
 r_{n-1} &= q_{n+1}r_n + 0.
 \end{aligned}$$

The first non-zero remainder is $r_n|\gcd(a, b)$, or a common divisor. To show that it's the greatest common divisor, suppose $d = as' + bt'$ for some $s', t' \in \mathbb{Z}$ where $d = \gcd(a, b)$. Then if $a = q_1b + r_1$ and $b = q_2r_1 + 0$, if $d > r_1$, but d is a common divisor, then $a - q_1b = r_1$. Thus $d|a, d|b \implies d|r_1$, therefor d can't be bigger than r_1 .

Example 4.5. Let $a = 23, b = 3$. Since $23 = 1 \cdot 3 + 20$, we have $q = 1$ and $r = 20$ which is not valid since r is not less than b . We also have $23 = 8 \cdot 3 - 1$, so $q = 8$ and $r = -1$ is not valid since r is not nonnegative.

The only valid one is $23 = 3 \cdot 7 + 2$, so $q = 7$ and $r = 2$. There is no other way to write $23 = 3 \cdot q + r$ where $0 \leq r < 3$.

Example 4.6. Use the Division Algorithm to find $\gcd(2520, 378)$.

$$\begin{aligned}
 2520 &= 378(6) + 252 \\
 378 &= 252(1) + 126 \\
 252 &= 126(2) + 0
 \end{aligned}$$

The last non-zero remainder is the gcd, so $\gcd(2520, 378) = 126$.

5 Lecture 5 – 1/28

Theorem 5.1. Let p be an integer such that $p > 1$. We say that p is a **prime number**, or simply p is **prime**, if the only positive divisors of p are 1 and p itself. An integer greater than 1 that is not prime is called a **composite number**.

Lemma. Euclid. Let a and b be integers and p be a prime number. If $p|ab$, then either $p|a$ or $p|b$. Suppose $p|ab$ and suppose $p \nmid a$. Then $\gcd(a, p) = 1$. Thus there exist integers s, t such that $as + pt = 1$. Multiplying both sides by b gives $abs + ptb = b$. Since $p|ab$ and $p|ptb$, we have $p|b$.

Theorem 5.2. There exist an infinite number of primes.

Assume $p_1, p_2, \dots, p_n$ are all the primes. Let $N = p_1 p_2 \cdots p_n + 1$. Then N is either prime or composite. If N is prime, then there exists a prime not in our list. If N is composite, then it has a prime divisor p . Since p divides N and p divides $p_1 p_2 \cdots p_n$, p must also divide their difference, which is 1. This is a contradiction since no prime divides 1. Thus there exists a prime not in our list.

5.1 3.1 Integer Equivalence Classes and Symmetries

$\mathbb{Z}_n = \{[0], [1], [2], \dots, [n-1]\}$ where arithmetic is mod n and the equivalence classes are $a \equiv b \pmod{n}$ if $n|a - b$.

Example 5.3. $\mathbb{Z}_6$:

$[0] = \{0, \pm 6, \pm 12, \pm 18 \cdots\} = [6] = [-12]$ all have a remainder of 0 when divided by 6.

$[1] = \{1, \pm 5, \pm 11, \pm 17 \cdots\} = [7] = [-11]$ all have a remainder of 1 when divided by 6.

$\vdots$

6 Lecture 6 – 1/29

Definition 6.1. Let $\mathbb{Z}_n$ be the set of equivalence classes of the integers *mod* n and $a, b, c \in \mathbb{Z}_n$.

1. Addition and multiplication are commutative. $ab \equiv ba$ and $a + b \equiv b + a$.
2. Addition and multiplication are associative. $(ab)c \equiv a(bc)$ and $(a + b) + c \equiv a + (b + c)$.
3. There are both additive and multiplicative identities. $0 \in \mathbb{Z}_n$ has the property $a + 0 \equiv a$ and $1 \in \mathbb{Z}_n$ has the property $a \cdot 1 \equiv a$.
4. Multiplication distributes over addition. $a(b + c) \equiv ab + ac$.
5. For every integer a there is an additive inverse $-a$. $a + (n - a) \equiv (n - a) + a \equiv n \equiv 0$.
6. Let a be a nonzero integer. Then $\gcd(a, n) = 1$ if and only if there exists a multiplicative b inverse for $a(\text{mod } n)$.

Definition 6.2. A **symmetry** of a geometric figure is a rearrangement of the figure preserving the arrangement of its sides and vertices as well as its distances and angles. A map from the plane to itself preserving the symmetry of an object is called a **rigid motion**.

6.1 3.2 Definitions and Examples

Definition 6.3. A **binary operation** or **law of composition** on a set G is a function $G \times G \rightarrow G$ that assigns to each pair $(a, b) \in G \times G$ a unique element $a \circ b$, or ab in G , called the composition of a and b .

Definition 6.4. A **group** is a set G together with a law of composition $\circ$ that satisfies the following axioms:

- The law of composition is **associative**; that is, for all $a, b, c \in G$, we have $(a \circ b) \circ c = a \circ (b \circ c)$.
- There exists an **identity element** e in G such that for all $a \in G$, we have $e \circ a = a \circ e = a$.
- For each element $a \in G$, there exists an **inverse element** $a^{-1} \in G$ such that $a \circ a^{-1} = a^{-1} \circ a = e$.

A group with the property that $a \circ b = b \circ a$ for all $a, b \in G$ is called an **abelian group** or **commutative group**. Groups not satisfying this property are called **nonabelian** or **non-commutative**.

7 Lecture 7 – 2/3

The identity element in a group G is unique; that is, there exists only one element $e \in G$ such that $eg = ge = g$ for all $g \in G$.

If g is any element in a group G , then the inverse of g , denoted by g^{-1} , is unique.

Let G be a group. If $a, b \in G$, then $(ab)^{-1} = b^{-1}a^{-1}$.

Let G be a group. For any $a \in G$, $(a^{-1})^{-1} = a$.

Let G be a group and a and b be any two elements in G . Then the equations $ax = b$ and $xa = b$ have unique solutions in G .

If G is a group and $a, b, c \in G$, then $ba = ca$ implies $b = c$ and $ab = ac$ implies $b = c$.

Theorem 7.1. In a group, the usual law of exponents hold; that is, for all $g, h \in G$,

1. $g^m g^n = g^{m+n}$ for all $m, n \in \mathbb{Z}$.
2. $(g^m)^n = g^{mn}$ for all $m, n \in \mathbb{Z}$.
3. $(gh)^n = (h^{-1}g^{-1})^{-n}$ for all $n \in \mathbb{Z}$. Furthermore, if G is abelian, then $(gh)^n = g^n h^n$.

8 Lecture 8 – 2/5

8.1 3.3 Subgroups

We define a **subgroup** H of a group G to be a subset H of G such that when the group operation of G is restricted to H , H is a group in its own right.

Observe that every group with at least two elements will always have at least two subgroups, the subgroup consisting of the identity element alone and the entire group itself. The subgroup $H = \{e\}$ of a group is called the **trivial subgroup**. A subgroup that is a proper subset of is called a **proper subgroup**.

Example 8.1. Let $G = (\mathbb{Z}, +)$, and $H = (\mathbb{Z}, +)$. H is a trivial subgroup of G . Then let $H' = (\{0\}, +)$. H' is the trivial subgroup of G .

Since $\mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$, we have: $(\mathbb{C}, +)$ is a group, $(\mathbb{R}, +)$ is a subgroup of $(\mathbb{C}, +)$. If $H = (\mathbb{R}, +)$ is a subgroup of $G = (\mathbb{C}, +)$, then consider $(a_1 + b_1 i) + (a_2 + b_2 i) \in G \implies (a_1 + a_2) + (b_1 + b_2)i \in G$.

Let H be a subset of group G . Then H is a subgroup of G if and only if

- **Closure:** For all $a, b \in H$, $ab \in H$.
- **Identity:** The identity element e of G is in H .
- **Inverses:** For all $a \in H$, $a^{-1} \in H$.

8.2 4.1 Cyclic Subgroups

Theorem 8.2. Let G be a group and a be any element in G . Then the set $\langle a \rangle = \{a^n : n \in \mathbb{Z}\}$ is a subgroup of G . Furthermore, $\langle a \rangle$ is the smallest subgroup of G that contains a and is called the **cyclic subgroup** of G generated by a .

9 Lecture 9 – 2/10

For $a \in G$, we call $\langle a \rangle$ the **cyclic subgroup** of G generated by a .

If G contains some element a such that $G = \langle a \rangle$, then G is a **cyclic group** and a is called a **generator** of G .

If a is an element of a group G , we define the **order** of a to be the smallest positive integer n such that $a^n = e$, and we write $|a| = n$. If there is no such integer n , we say that the order of a is infinite and write $|a| = \infty$ to denote the order of a .

Theorem 9.1. Every cyclic group is abelian.

Also, every subgroup of a cyclic group is cyclic.

Corollary: The subgroups of $\mathbb{Z}$ are exactly $n\mathbb{Z}$ for $n = 0, 1, 2, \dots$.

10 Lecture 10 – 2/11

Proposition: Let G be a cyclic group of order n and suppose that a is a generator for G . Then $a^k = e$ if and only if $n|k$, or k divides n .

Theorem 10.1. Let G be a cyclic group of order n and suppose that a is a generator of the group. If $b = a^k$, then the order of b is $\frac{n}{d}$, where $d = \gcd(k, n)$.

Corollary: The generators of $\mathbb{Z}_n$ are the integers r such that $1 \leq r < n$ and $\gcd(r, n) = 1$.

11 Lecture 11 – 2/12

11.1 4.2 Multiplicative Group of Complex Numbers

$\mathbb{C} = \{x + iy : x, y \in \mathbb{R}, i^2 = -1\}$.

If $z = x + iy$, then $z = r(\cos \theta + i \sin \theta) = cis\theta = re^{i\theta}$ where $r = \sqrt{x^2 + y^2}$ and $\theta = \tan^{-1} \frac{y}{x}$.

The circle group $\mathbb{T} = \{z \in \mathbb{C} : |z| = 1\}$ is a subgroup of $\mathbb{C}^*$.

Theorem 11.1. If $z^n = 1$, then the n^{th} roots of unity are $z = \cos \frac{2k\pi}{n} + i \sin \frac{2k\pi}{n}$ for $k = 0, 1, 2, \dots, n-1$. Furthermore, the n^{th} roots of unity form a cyclic subgroup of $\mathbb{T}$ of order n .

11.2 5.1 Definitions and Notation

Theorem 11.2. The symmetric group on n letters, S_n , is a group with $n!$ elements, where the binary operation is the composition of maps.

Cycle Notation: A permutation $\sigma \in S_X$ is a cycle of length k if there exist elements $a_1, a_2, \dots, a_k \in X$ such that $\sigma(a_1) = a_2, \sigma(a_2) = a_3, \dots, \sigma(a_k) = a_1$, and $\sigma(x) = x$ for all other $x \in X$. We write $(a_1 a_2 \dots a_k)$ to denote the cycle σ .

A subgroup of S_n is called a **permutation group**.

Two cycles in S_n , $\sigma = (a_1 a_2 \dots a_k)$ and $\tau = (b_1 b_2 \dots b_m)$ are **disjoint** if $a_i \neq b_j$ for all i and j .

Let σ and τ be two disjoint cycles in S_n . Then $\sigma\tau = \tau\sigma$.

Every permutation in S_n can be written as the product of disjoint cycles.

12 Lecture 12 – 2/18

- A cycle of length 2 is called a **transposition**.
- **Proposition:** Any permutation of a finite set containing at least two elements can be written as the product of transpositions.
- If the identity id is written as the product of r transpositions, $id = \tau_1 \tau_2 \dots \tau_r$, then r is even.

Theorem 12.1. If a permutation σ can be expressed as the product of an even number of transpositions, then any other product of transpositions equaling must also contain an even number of transpositions.

We define a permutation to be even if it can be expressed as an even number of transpositions and odd if it can be expressed as an odd number of transpositions.

Theorem 12.2. The set of even permutations A_n is a subgroup of S_n .

Proposition: The number of even permutations is $S_n, n \geq 2$, is equal to the number of odd permutations; hence the order of A_n is $\frac{n!}{2}$.

13 Lecture 13 – 2/19

13.1 5.2 Dihedral Groups

For $n = 3, 4, \dots$, we define the **n th dihedral group** to be the group of rigid motions of a regular n -gon.

Theorem 13.1. The dihedral group, D_n , is a subgroup of S_n of order $2n$.

Theorem 13.2. The group $D_n, n \geq 3$, consists of all products of the elements r and s , where r has order n and s has order 2, and these two elements satisfy the relation $srs = r^{-1}$.

13.2 6.1 Cosets

Let G be a group and H be a subgroup of G . Define a **left coset** of H with representative $g \in G$ to be the set $gH = \{gh : h \in H\}$. Define a **right coset** of H with representative g to be the set $Hg = \{hg : h \in H\}$.

14 Lecture 14 – 2/26

Lemma: Let H be a subgroup of a group G and suppose that $g_1, g_2 \in G$. The following conditions are equivalent:

- $g_1H = g_2H$
- $Hg_1^{-1} = Hg_2^{-1}$
- $g_1H \subset g_2H$
- $g_2 \in g_1H$
- $g_1^{-1}g_2 \in H$.

Theorem 14.1. Let H be a subgroup of a group G . Then the left cosets of H in G form a partition of G . That is, the group G is the disjoint union of the left cosets of H in G .

14.1 6.2 Lagrange's Theorem

Proposition: Let H be a subgroup of G with $g \in G$ and define a map $\phi : H \rightarrow gH$ by $\phi(h) = gh$. The map is bijective; hence, the number of elements in H is the same as the number of elements in gH .

Theorem 14.2. Let G be a finite group and let H be a subgroup of G . Then $|G|/|H| = [G : H]$ is the number of distinct left cosets of H in G , called the **index** of H in G . In particular, the number of elements in H must divide the number of elements in G .

Corollary: Suppose that G is a finite group and $g \in G$. Then the order of g must divide the number of elements in G .

Corollary: Let $|G| = p$ with p a prime number. Then G is cyclic and any $g \in G$ such that $g \neq e$ is a generator.

Corollary: Let H and K be subgroups of a finite group G such that $K \subset H \subset G$. Then $[G : K] = [G : H][H : K]$.

14.2 6.3 Fermat's and Euler's Theorems

The **Euler ϕ -function** is the map $\phi : \mathbb{N} \rightarrow \mathbb{N}$ defined by $\phi(1) = 1$, and, for $n > 1$, $\phi(n)$ is the number of positive integers m with $1 \leq m < n$ and $\gcd(m, n) = 1$.

15 Lecture 15 – 3/3

Theorem 15.1. Let $U(n)$ be the group of units in $\mathbb{Z}_n$. Then $|U(n)| = \phi(n)$.

Theorem 15.2. Euler's theorem: Let a and n be integers such that $n > 0$ and $\gcd(a, n) = 1$. Then $a^{\phi(n)} \equiv 1 \pmod{n}$.

Example 15.3. What is the last digit of 457^{327} ?

Since $457 \equiv 7 \pmod{10} \implies 457^{327} \equiv 7^{327} \pmod{10}$. Since $\gcd(7, 10) = 1$, we can apply Euler's theorem to get $7^{\phi(10)} \equiv 1 \pmod{10}$. Since $\phi(10) = 4$, we have $7^4 \equiv 1 \pmod{10}$. Thus $7^{327} \equiv 7^{4 \cdot 81 + 3} \equiv (7^4)^{81} \cdot 7^3 \equiv 1^{81} \cdot 343 \equiv 3 \pmod{10}$. Therefore the last digit of 457^{327} is 3.

Theorem 15.4. Fermat's Little Theorem: Let p be any prime number and suppose that $p \nmid a$ (p does not divide a). Then $a^{p-1} \equiv 1 \pmod{p}$. Furthermore, for any integer b , $b^p \equiv b \pmod{p}$.

15.1 9.1 Isomorphisms: Definition and Examples

Two groups $(G, \cdot)$ and $(H, \circ)$ are **isomorphic** if there exists a one-to-one and onto map $\phi : G \rightarrow H$ such that the group operation is preserved; that is, $\phi(a \cdot b) = \phi(a) \circ \phi(b)$ for all $a, b \in G$. If G is isomorphic to H , we write $G \cong H$. The map is called an isomorphism.

16 Lecture 16 – 3/4

Theorem 16.1. All cyclic groups of infinite order are isomorphic to $\mathbb{Z}$.

Theorem 16.2. If G is a cyclic group of order n , then G is isomorphic to $\mathbb{Z}_n$.

Corollary: If G is a group of order p , where p is a prime number, then G is isomorphic to $\mathbb{Z}_p$.

Theorem 16.3. Cayley: Every group is isomorphic to a group of permutations.

16.1 9.2 Direct Products

If $(G, \cdot)$ and $(H, \circ)$ are groups, then we can make the Cartesian product of G and H into a new group. As a set, our group is just the ordered pairs $(g, h) \in G \times H$. We can define a binary operation on $G \times H$ by $(g_1, h_1)(g_2, h_2) = (g_1 \cdot g_2, h_1 \circ h_2)$.

Proposition: Let G and H be groups. The set $G \times H$ is a group. The group $G \times H$ is called the external direct product of G and H .

Theorem 16.4. Let $(g, h) \in G \times H$. If g and h have finite orders r and s respectively, then the order of (g, h) in $G \times H$ is the least common multiple of r and s .

Theorem 16.5. The group $\mathbb{Z}_m \times \mathbb{Z}_n$ is isomorphic to $\mathbb{Z}_{mn}$ if and only if $\gcd(m, n) = 1$.

17 Lecture 17 – 3/5

Corollary: Let $n_1, \dots, n_k$ be positive integers. Then

$$\prod_{i=1}^k \mathbb{Z}_{n_i} \cong \mathbb{Z}_{n_1 \cdots n_k}$$

if and only if $\gcd(n_i, n_j) = 1$ for $i \neq j$.

Corollary: If $m = p_1^{e_1} \cdots p_k^{e_k}$ where the p_i s are distinct primes, then

$$\mathbb{Z}_m \cong \mathbb{Z}_{p_1^{e_1}} \times \cdots \times \mathbb{Z}_{p_k^{e_k}}.$$

Let G be a group with subgroups H and K satisfying the following conditions:

- $G = HK = \{hk : h \in H, k \in K\}$,
- $H \cap K = \{e\}$,
- $hk = kh$ for all $h \in H$ and $k \in K$.

Then G is the **internal direct product** of H and K .

Theorem 17.1. Let G be the internal direct product of subgroups H and K . Then G is isomorphic to $H \times K$.

18 Lecture 17 – 3/17

18.1 10.1 Factor Groups and Normal Subgroups

A subgroup H of a group G is **normal** in G if $gH = Hg$ for all $g \in G$.

Example 18.1. Consider $3\mathbb{Z}$ in $\mathbb{Z}$.

$$H = 3\mathbb{Z} = \{\dots, -6, -3, 0, 3, 6, \dots\} = \{3n : n \in \mathbb{Z}\}.$$

$$0 + H = 3 + H = 6 + H = \dots = H + 0 = H + 3 = \dots \quad 1 + H = 4 + H = 7 + H = \dots = H + 1 = H + 4 = \dots \quad 2 + H = 5 + H = 8 + H = \dots = H + 2 = H + 5 = \dots$$

If G is abelian, then every subgroup is normal.

Example 18.2. Consider $H = \{e, (12)\}$ in S_3 .

$$\text{Note } eH = He = H \text{ and } (12)H = \{(12), e\} = \{e, (12)\} = H(12).$$

$$\text{Consider } (123)H = \{(123)e, (123)(12)\} = \{(123), (13)\} \text{ and } H(123) = \{e(123), (12)(123)\} = \{(123), (23)\}.$$

$$H(123) = \{(123), (23)\} \neq \{(123), (13)\} = (123)H. \text{ Thus } H \text{ is not normal in } S_3.$$

Example 18.3. Consider A_3 in S_3 .

$$A_3 = \{e, (123), (132)\}. \text{ We know that } eA_3 = A_3e = A_3. \text{ Also, } (123)A_3 = \{(123), (132), e\} = A_3(123) \text{ and } (132)A_3 = \{(132), e, (123)\} = A_3(132).$$

$$\text{Consider } (12)A_3 = \{(12), (13), (23)\} \text{ and } A_3(12) = \{(12), (13), (23)\}.$$

$$\text{So } (12)A_3 = A_3(12), \text{ and } A_3 \text{ is normal in } S_3.$$

Theorem 18.4. Let G be a group and N be a subgroup of G . Then the following statements are equivalent.

- The subgroup N is normal in G .
- For all $g \in G$, $gNg^{-1} = N$.
- For all $g \in G$, $gNg^{-1} \subset N$.

Definition 18.5. If N is a normal subgroup of a group G , then the cosets of N in G form a group G/N under the operation $(aN)(bN) = abN$. This group is called the **factor** or **quotient group** of G and N .

Example 18.6. What is the factor group $\mathbb{Z}/3\mathbb{Z}$?

$$\mathbb{Z}/3\mathbb{Z} = \{0 + 3\mathbb{Z}, 1 + 3\mathbb{Z}, 2 + 3\mathbb{Z}\} =$$

$$\{\{\dots, -6, -3, 0, 3, 6, \dots\}, \{\dots, -5, -2, 1, 4, 7, \dots\}, \{\dots, -4, -1, 2, 5, 8, \dots\}\}.$$

$$(0 + 3\mathbb{Z}) + (1 + 3\mathbb{Z}) = (0 + 1) + 3\mathbb{Z} = 1 + 3\mathbb{Z}.$$

$$(1 + 3\mathbb{Z}) + (2 + 3\mathbb{Z}) = (1 + 2) + 3\mathbb{Z} = 0 + 3\mathbb{Z}.$$

$$(2 + 3\mathbb{Z}) + (2 + 3\mathbb{Z}) = (2 + 2) + 3\mathbb{Z} = 1 + 3\mathbb{Z}.$$

Theorem 18.7. Let N be a normal subgroup of a group G . The cosets of N in G form a group G/N of order $[G : N]$.

19 Lecture 18 – 3/18

19.1 11.1 Group Homomorphisms

A **homomorphism** between groups $(G, \cdot)$ and $(H, \circ)$ is a map $\phi : G \rightarrow H$ such that $\phi(g_1 \cdot g_2) = \phi(g_1) \circ \phi(g_2)$ for $g_1, g_2 \in G$. The range of ϕ is called the **homomorphic image** of G .

Example 19.1. Let $\phi : \mathbb{Z} \rightarrow \mathbb{Z}_4 \times \mathbb{Z}_9$ be a homomorphism with $\phi(1) = (2, 3)$. Find $\phi(n)$ for any $n \in \mathbb{Z}$. Which elements $n \in \mathbb{Z}$ have $\phi(n) = (0, 0)$?

$$\phi(2) = \phi(1 + 1) = \phi(1) + \phi(1) = (2, 3) + (2, 3) = (0, 6) \in \mathbb{Z}_4 \times \mathbb{Z}_9.$$

$$\phi(3) = \phi(1) + \phi(1) + \phi(1) = \phi(2) + \phi(1) = (0, 6) + (2, 3) = (2, 0).$$

$$\phi(4) = (0, 3), \phi(5) = (2, 6), \phi(6) = (0, 0).$$

Let $\phi : G_1 \rightarrow G_2$ be a homomorphism of groups. Then

1. If e is the identity of G_1 , then $\phi(e)$ is the identity of G_2 .
2. For any element $g \in G$, $\phi(g^{-1}) = [\phi(g)]^{-1}$.
3. If H_1 is a subgroup of G_1 , then $\phi(H_1)$ is a subgroup of G_2 .
4. If H_2 is a subgroup of G_2 , then $\phi^{-1}(H_2) = \{g \in G_1 : \phi(g) \in H_2\}$ is a subgroup of G_1 ;
Furthermore, if H_2 is normal in G_2 , then $\phi^{-1}(H_2)$ is normal in G_1 .

Let $\phi : G \rightarrow H$ be a group homomorphism and suppose that e is the identity of H . The subgroup $\phi^{-1}(\{e\})$ is called the **kernel** of ϕ and is denoted by $\ker(\phi)$.

Theorem 19.2. Let $\phi : G \rightarrow H$ be a group homomorphism. Then the kernel of ϕ is a normal subgroup of G .

20 Lecture 19 – 3/19

20.1 11.2 The Isomorphism Theorems

Let H be a normal subgroup of G . Define the **natural** or **canonical homomorphism** $\phi : G \rightarrow G/H$ by $\phi(g) = gH$.

Example 20.1. Let $G = \mathbb{Z}$ and $H = 5\mathbb{Z}$.

Let $\phi : \mathbb{Z} \rightarrow \mathbb{Z}/5\mathbb{Z}$, $\phi(n) = n + 5\mathbb{Z}$.

$$\phi(0) = 0 + 5\mathbb{Z} = [0], \phi(1) = 1 + 5\mathbb{Z} = [1], \phi(2) = 2 + 5\mathbb{Z} = [2], \phi(3) = 3 + 5\mathbb{Z} = [3],$$

$$\phi(4) = 4 + 5\mathbb{Z} = [4], \phi(5) = 0 + 5\mathbb{Z} = [0]. \text{ Note: } \ker(\phi) = 5\mathbb{Z} \text{ (kernel: a group and happens to be the image of } \phi).$$

Theorem 20.2. First Isomorphism Theorem: If $\psi : G \rightarrow H$ is a group homomorphism with $K = \ker\psi$, then K is normal in G . Let $\phi : G \rightarrow G/K$ be the canonical homomorphism. Then there exists a unique isomorphism $\eta : G/K \rightarrow \psi(G)$ such that $\psi = \eta\phi$.

Example 20.3. Let $\psi : \mathbb{Z} \rightarrow \mathbb{Z}_6$ be a group homomorphism with $\psi(1) = 2$. Verify the First Isomorphism Theorem.

$$\psi(n) = 2n \pmod{6}. K = \ker\psi = 3\mathbb{Z} \subset \mathbb{Z}.$$

Let $\phi : \mathbb{Z} \rightarrow \mathbb{Z}/\ker\psi \implies \phi : \mathbb{Z} \rightarrow \mathbb{Z}/3\mathbb{Z}$ be the canonical homomorphism. Then the First Iso. Theorem $\implies \mathbb{Z}/3\mathbb{Z} \cong \text{image of } \psi$.

Theorem 20.4. Second Isomorphism Theorem: Let H be a subgroup of a group G (not necessarily normal in G) and N a normal subgroup of G . Then HN is a subgroup of G , $H \cap N$ is a normal subgroup of H , and $HN/N \cong H/H \cap N$.

Example 20.5. Let $H = 6\mathbb{Z}$ and $N = 10\mathbb{Z}$ be subgroups of $G = \mathbb{Z}$. Verify the Second Isomorphism Theorem.

$$HN = \{h + n : h \in H, n \in N\} \implies (6\mathbb{Z})(10\mathbb{Z}) = \{(6m) + (10n) : m, n \in \mathbb{Z}\} = 2\mathbb{Z}.$$

$$H \cap N = \{m : m = 6k, m = 10n \text{ for some } k, n \in \mathbb{Z}\} = 30\mathbb{Z}.$$

$$H/H \cap N = 6\mathbb{Z}/30\mathbb{Z} = \{[0], [6], [12], [18], [24]\}.$$

$$HN/N = 2\mathbb{Z}/10\mathbb{Z} = \{[0], [2], [4], [6], [8]\}.$$

21 Lecture 20 – 3/24

Theorem 21.1. Correspondence Theorem: Let N be a normal subgroup of a group G . Then $H \mapsto H/N$ is a one-to-one correspondence between the set of subgroups H of G containing N and the set of subgroups of G/N . Furthermore, the normal subgroups of G containing N correspond to normal subgroups of G/N .

Theorem 21.2. Third Isomorphism Theorem: Let G be a group and N and H be normal subgroups of G with $N \subset H$. Then $G/H \cong \frac{G/N}{H/N}$.

Example 21.3. Let $G = \mathbb{Z}$, $H = 4\mathbb{Z}$, and $N = 24\mathbb{Z}$. Verify the Third Isomorphism Theorem.

$$24\mathbb{Z} = N \subset 4\mathbb{Z} = H \subset \mathbb{Z} = G.$$

$$G/H = \mathbb{Z}/4\mathbb{Z} \cong \mathbb{Z}_4.$$

$$G/N = \mathbb{Z}/24\mathbb{Z} \cong \mathbb{Z}_{24}.$$

$$H/N = 4\mathbb{Z}/24\mathbb{Z} \cong \mathbb{Z}_6.$$

$$\frac{G/N}{H/N} = \frac{\mathbb{Z}/24\mathbb{Z}}{4\mathbb{Z}/24\mathbb{Z}} \cong \mathbb{Z}_4.$$